

Examination paper for TDT4175 Information Systems

Academic contact during examination: Pieter Toussaint and Patrick Mikalef

Phone: 40646586 or 48502737

Examination date: 16 December 2019

Examination time (from-to): 9:00 – 13:00

Permitted examination support material: Allowed aids: C Specified printed and handwritten material is allowed. In the run of the course the book "BPMN Method & Style", Bruce Silver. (Edition 1 or 2) has been specified. Clean copies of the chapters 1-6 used in the course also allowed. In addition standard dictionaries (English-English, and from English to other languages) are allowed to bring. No additional printed or handwritten materials are allowed. An approved simple calculator is allowed.

Language: English

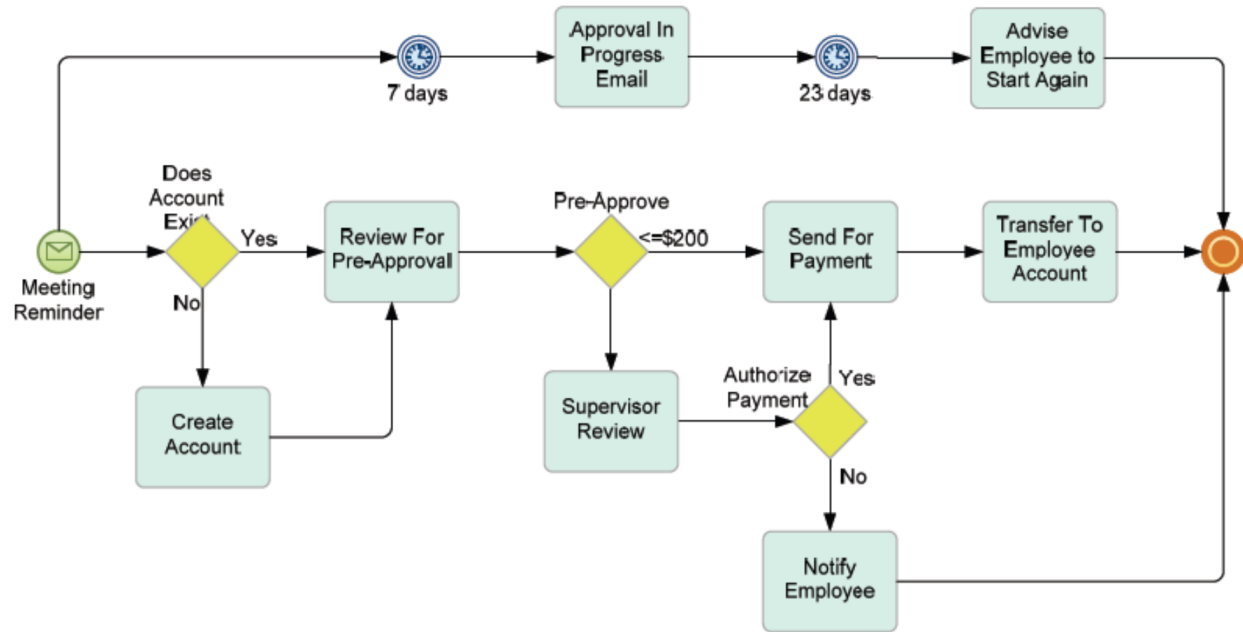
Number of pages (front pages excluded): 2

Number of pages enclosed: 3

Task 1 (20 points) - Design a sample expense reimbursement process using BPMN. This process provides for reimbursement of expenses incurred by employees for the company in which they work. Examples of reimbursements include: buying a technical book, office supplies or software etc. This process is handled by the financial department.

After the reception of a reimbursement claim, a new account must be created if the employee does not already have one. The claim is then reviewed for automatic approval. Amounts under \$200 are automatically approved, whereas amounts equal to or over \$200 require approval of the supervisor. In case of rejection, the employee must receive a rejection notice by email. The reimbursement goes to the employee's direct deposit bank account if it is automatically accepted or is accepted by the supervisor. If the request is not completed in 7 days, then the employee must receive an "approval in progress" email. If

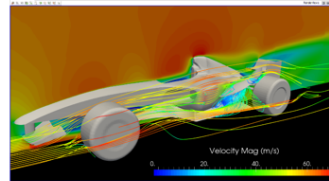
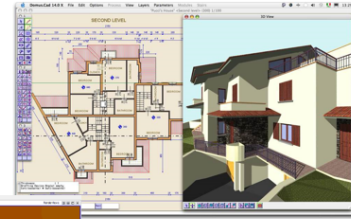
the request is not finished within 30 days, then the process is stopped and the employee is sent an email to start the process again.



Task 2 (5 points) – List the components of a Product Lifecycle Management (PLM) system and briefly describe each of them and their functions.

Product Lifecycle Management (PLM)

- Computer-aided design (CAD):
 - ✓ The use of software to assist in the creation, analysis, and modification of the design of a component or product
- Computer-aided engineering (CAE):
 - ✓ The use of software to analyze the robustness and performance of components and assemblies
- Computer-aided manufacturing (CAM)
 - ✓ The use of software to control machine tools and related machinery in the manufacture of components and products



Task 3 (5 points) – In Lecture 3 we discussed different types of enterprise systems. Which type of system would you use to support the processes from task 1. Motivate your choice.

The choice of a Transaction Processing System (TPS) would be the best fitting here as the case described basic transactions that can be automated.

Task 4 (5 points) – List the stages in Lewin's Change Model and briefly outline what each stage includes.

Lewin's Change Model

- Lewin's change model consists of a three-stage approach for change

1. **Unfreezing**: preparing for change



2. **Moving**: making the change



3. **Refreezing**: institutionalizing



Task 5 (10 points) – In Lecture 2 we talked about Miles and Snow’s strategy typology. List the different strategies that they distinguish **(4 of 10 points)**. In which typology would you place Google and why? Motivate your answer **(6 of 10 points)**.

13

Miles and Snow’s Strategy Typology

Strategy Type	Definition	Examples
Prospector	In innovative and growth oriented, searches for new markets and new growth opportunities, encourages risk taking	Amazon Google
Defender	Protects current markets, maintains stable growth, serves current customers	BIC Corporation YKK Group
Analyzer	Maintains current markets and current customer satisfaction with moderate emphasis on innovation	IBM Yahoo UBER
Reactor	No clear strategy, reacts to changes in the environments, drifts with events	NOKIA

Task 6 (5 points) – In Lecture 1 we talked about the dimensions of valuable information. List 3 that you believe are relevant for the case of credit card transactions and motivate your choice by briefly describing why they are important.

Characteristics	Definitions
Accessible	Information should be easily accessible by authorized users so they can obtain it in the right format and at the right time to meet their needs.
Accurate	Accurate information is error free. In some cases, inaccurate information is generated because inaccurate data is fed into the transformation process. (This is commonly called garbage in, garbage out [GIGO].)
Complete	Complete information contains all the important facts. For example, an investment report that does not include all important costs is not complete.
Economical	Information should also be relatively economical to produce. Decision makers must always balance the value of information with the cost of producing it.
Flexible	Flexible information can be used for a variety of purposes. For example, information on how much inventory is on hand for a particular part can be used by a sales representative in closing a sale, by a production manager to determine whether more inventory is needed, and by a financial executive to determine the total value the company has invested in inventory.
Relevant	Relevant information is important to the decision maker. Information showing that lumber prices might drop might not be relevant to a computer chip manufacturer.
Reliable	Reliable information can be trusted by users. In many cases, the reliability of the information depends on the reliability of the data-collection method. In other instances, reliability depends on the source of the information. A rumor from an unknown source that oil prices might go up might not be reliable.
Secure	Information should be secure from access by unauthorized users.
Simple	Information should be simple, not overly complex. Sophisticated and detailed information might not be needed. In fact, too much information can cause information overload, whereby a decision maker has too much information and is unable to determine what is really important.
Timely	Timely information is delivered when it is needed. Knowing last week's weather conditions will not help when trying to decide what coat to wear today.
Verifiable	Information should be verifiable. This means that you can check it to make sure it is correct, perhaps by checking many sources for the same information.

Table 1.2

Characteristics of Valuable Information

Some potential relevant would be: Accurate, Secure, Timely, Verifiable, but also others if they are justified.

Task 7 (5 points) – In Lecture 3 we discussed different types of e-commerce. List 3 of these and for each give a real-life example.

E-COMMERCE ACTIVITIES



B2C

Business to Consumer

When person buys products or services online.



B2B

Business to Business

When E-Commerce Website orders from suppliers or sellers to deliver to customer.



C2C

Consumer to Consumer

When Consumers directly trade with one another.



G2B G2C

Government to Business or Consumer

When government transacts with business or consumer online.

Task 8 (10 points) – You have been requested to digitalize the processes of a local manufacturing company so that there is greater control of inventory, finances, customers and sales and marketing. Which type of Information System would you use to do that? Motivate your answer.



An ERP would be a good choice due to the modules that can be used to fulfil the different requirements of the manufacturing company.

Task 9 (10 points) – You have been asked to suggest an information system for a hospital so that doctors can share their best practices and experiences with colleagues. What type of system is best suited for such a use? Describe specifically which technology you would use within this type of information systems and motivate your choice.

The answer should be focused on a Knowledge Management System due to the need to codify knowledge. Several alternative answers can be accepted in the use of the specific technology as long as they are well-motivated. For instance, expert systems, communities of practice, forums, or other web 2.0 tools.

Task 10 (5 points) – What is the difference between re-engineering and continuous improvement?

Re-Engineering and Continuous Improvement

- Re-engineering
 - ✓ Also called process redesign and business process reengineering (BPR)
 - ✓ Involves the **radical** redesign of business processes, organizational structures, information systems, and values of the organization to achieve a breakthrough in business results
- Continuous improvement
 - ✓ Constantly seeking ways to **improve** business processes and add value to products and services